

Natural Language Processing Reveals Core Issues in Uganda's Power Grid: A Study of Outages from 2015-2022

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Abstract—This research study investigates the root causes of power outages in Uganda, leveraging a unique data set obtained through a partnership with the country's electricity regulator. Covering the period from 2015 to 2022, this data set includes detailed records of each power outage, noting the duration and free-form descriptions of the incidents. The researchers used an advanced natural language processing (NLP) tool to categorize these descriptions into specific root cause groups. The findings highlight equipment failure as the most prevalent cause of power outages, suggesting a critical need to enhance or re-evaluate maintenance efforts. Furthermore, the study revealed that outages resulting from vandalism and fires tend to have significantly longer resolution times, indicating areas where targeted interventions could improve response efficiency. This research sheds light on the main factors contributing to power outages in Uganda and demonstrates how NLP can be applied to quickly and accurately identify these causes. The insights gained from this study offer valuable guidance for policymakers and utility companies aiming to improve the reliability and resilience of Uganda's power grid.

Index Terms—Reliability, Power Outages, Uganda, Umeme

I. INTRODUCTION

Power outages pose a significant challenge in many Sub-Saharan African countries, and Uganda is no exception. Historical data indicates a persistent struggle with unreliable electricity supply [1]–[3]. Numerous studies suggest these outages have far-reaching economic implications. For instance, Cole et al(2018) discovered a detrimental effect on firms' sales in countries with unreliable electricity, particularly those lacking backup generators [4]. Similarly, Abotsi et al(2016) found that frequent power outages diminish firms' production efficiency in Africa [5]. Andersen et al(2013) estimated that if all African nations matched South Africa's power quality, the continent's average annual real GDP per capita growth could increase by 2 percentage points [6]. Other studies corroborate these findings, emphasizing the broad economic and business repercussions of power outages [7]–[9].

Beyond the economic impacts, power outages reduce household leisure time and increase insecurity [10]. They also

diminish the time available for reading and work, thereby impacting education and overall productivity. Additionally, power outages force households to rely more on wood fuel and other dirty energy sources, which negatively affects health and socio-economic status [11]. Furthermore, power outages discourage households from purchasing certain appliances, further reducing electricity consumption levels, which are already very low in East African countries like Uganda [12], [13].

Understanding the root causes of power outages is crucial in addressing this pressing issue. The research by Amadi et al(2023) in Port Harcourt, Nigeria, identified inadequate power generating capacity, gas shortages, and infrastructure deficiencies as primary contributors to outages [14]. As highlighted by Rentschler et al(2019), urban flooding emerged as a significant cause of outages in Dar es Salaam, Tanzania. [7]. Meles (2020) suggests the main causes of power outages in Ethiopia are low capacity of the transmission and distribution lines, aged and poor physical conditions of transformers, and wastage of power due to high loss along these lines [15]. Neuhoff and Karsten (2020) attribute the persistent power outages in South Africa to generation constraints caused by aging and poorly maintained coal plants and high-price coal supply contracts [16]. However, Salman et al (2023) underscored a lack of comprehensive studies dedicated towards understanding power outage root causes, especially in developing nations [17]. This study aims to contribute to this knowledge gap by analyzing Uganda's power outages from 2015 to 2022. Leveraging a proprietary data set obtained from the Electricity Regulatory Authority (ERA), Uganda's electricity sector regulator, we employ a natural language processing model, SetFit, to analyze outage descriptions provided by technicians tasked with resolving power disruptions.

This investigation sheds light on the underlying causes of power outages in Uganda and lays the groundwork for

targeted interventions to improve the reliability and resilience of the country's electricity supply.

II. DATA

A. Outage dataset

Umeme, Uganda's primary distribution utility, is the source of the data we analyze. Our dataset is comprised of 319 unique feeders distributed across Umeme's service territory, as illustrated in Figure 1. The colored regions are the various service regions whose grid operation is maintained by Umeme. Kampala, the capital city, is divided into two service regions: Kampala East and Kampala West. While our dataset accounts for a total customer count of 1,007,118, this number is slightly lower than the 1.6 million reported by Umeme [18]. We attribute this discrepancy to potential missing records within our dataset, a challenge stemming from the utility's struggle to maintain up-to-date records of individual customer feeder assignments.

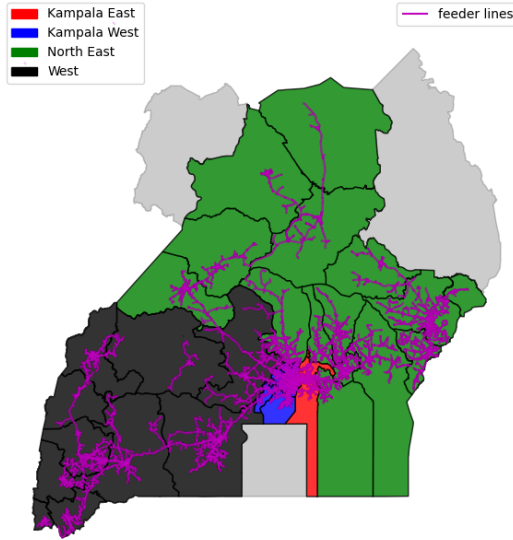


Fig. 1. Umeme service area

The data set spans eight years, from January 2015 to October 2022, and includes daily outage records. Each feeder is identified by a feeder code comprising the originating substation name, feeder name, feeder voltage level, and feeder count. For example, 'mtd/kbs/11/1' denotes a feeder originating from the Mutundwe substation (mtd), named Kabusu (kbs), operating at 11 kV voltage level, and designated as the first feeder on this line.

We utilize the recorded interruption time to identify the onset of an outage and calculate its duration by subtracting the restoration time. This approach allows us to analyze the frequency and duration of outages, our primary metrics for outage analysis.

The distribution of grid infrastructure and customers across Umeme's service territory is detailed in Table I, highlighting statistics for various regions. This breakdown includes the number of feeders, districts, and customers served within each region.

TABLE I
INFRASTRUCTURE AND CUSTOMER STATISTICS BY REGION

Regions	North East	West	KLA East	KLA West
No of feeders	65	49	119	86
No. of Districts	11	9	8	7
No. of Customers	126,102	196,488	412,500	272,028

B. Outage cause descriptions

For each power outage, technicians assign a type and provide a description. The type categorizes the outage as planned, unplanned, or an emergency shutdown, each defined as follows:

- **Planned Outages:** Scheduled electricity supply interruptions for maintenance, repair, upgrades, or other pre-planned activities on the power grid or related infrastructure.
- **Unplanned Outages:** Unscheduled electricity supply interruptions caused by unforeseen events or issues on the power grid or associated equipment.
- **Emergency Outages:** A subset of unplanned outages due to urgent or critical situations posing immediate risks to public safety, infrastructure, or the environment.

Technicians also provide a free-form description of the outage, such as its root cause, location, and resolution steps. Descriptions vary in detail; some outline the causes and rectification steps, while others confirm the outage's resolution. Sample entries are illustrated in Table II.

Periodically, ERA manually classifies these descriptions into 18 distinct classes that define the outage's root cause. We retain these 18 classes for our analysis but group them into broader categories to simplify the analysis process. From this point on, we refer to these broader groupings as categories and the original 18 classes as sub-categories. Figure 2 illustrates this categorization process. First, the power outage type is labeled as planned, unplanned, or emergency shutdowns, then labeled as one of the seven categories, and lastly, subdivided into one of the 18 sub-categories maintained by ERA's analytical teams.

III. METHODOLOGY

A. The SetFit Model

SetFit [19], a framework designed for few-shot fine-tuning of sentence transformers, has gained recognition for its capability to achieve remarkable accuracy even when confronted with a scarcity of labeled data. Thus rendering it a well-suited solution for scenarios characterized by sparse annotation.

TABLE II
EXAMPLE DESCRIPTIONS FROM DATASET

Description	Sub-category
broken jumper at kirigime	equipment accessories
Fault reclosure ssfull.	Unknown
UETCL works. Restored ssfully	Third Party
Broken pole at Kadetuni village	Pole Related
fault, local rec failed. Tree found knocked and fell on line at main hospital, fault cleared, rec ssf.	Vegetation
To replace a rotten pole at Mile 11 on East Rise road. Opned ABS at Mile 9. (PTW:49965).	pole related
Isolation of Ntake bakery faulted cable on Nalukolongo fdr.	cable related
Fault,reclosure failed,a bird fell on the line at Industrial area	animal
Fault, patrol still ongoing. Earth wire touching Cond. removed at Buhesi.	aerial earth wire
SHUTDOWN ON NORTHERN LINE (UETCL MTNCE WORKS)	third party
Kafumbe Mukasa transformer isolated due to burnt wirings, being rewired	transformer related
installing a metering unit on Bugweri rd for Industrial millers, open point at Bugwere rd	meter related
Fault, Tx vandalised on Entebbe at kisubi.	vandalism
Fault due to heavy rains, reclosure successful	weather
Loadshedded to improve poor voltage due to interconnector performance.	power quality

SetFit operates by fine-tuning a pre-trained sentence transformer on a small set of text pairs, utilizing few-shot learning techniques when training data is limited. Unlike conventional Neural Networks, which often require abundant examples for optimal performance, a pre-trained model like SetFit, trained on a substantial corpus of data, can be fine-tuned with limited data to effectively address specific categorization tasks, achieving high accuracy with little labeled data.

Figure 3 provides a sample code snippet demonstrating using SetFit. The code shows how to import the SetFit trainer, which inherits from the Hugging Face trainer [20]. This inheritance allows SetFit to be customizable and to utilize all the parameters available in the Hugging Face trainer.

SetFit supports loading data sets by directly passing the data set or providing file paths. Figure 3 illustrates a straightforward example of loading and training SetFit. The data provided consists of text and label fields used by the SetFit model in training and model validation.

We chose to utilize the SetFit model based on its superior performance demonstrated in previous research studies compared to other NLP approaches [21], [22]. SetFit is a state-of-the-art model with a reasonable model size and substantial support from a large community. Specifically de-

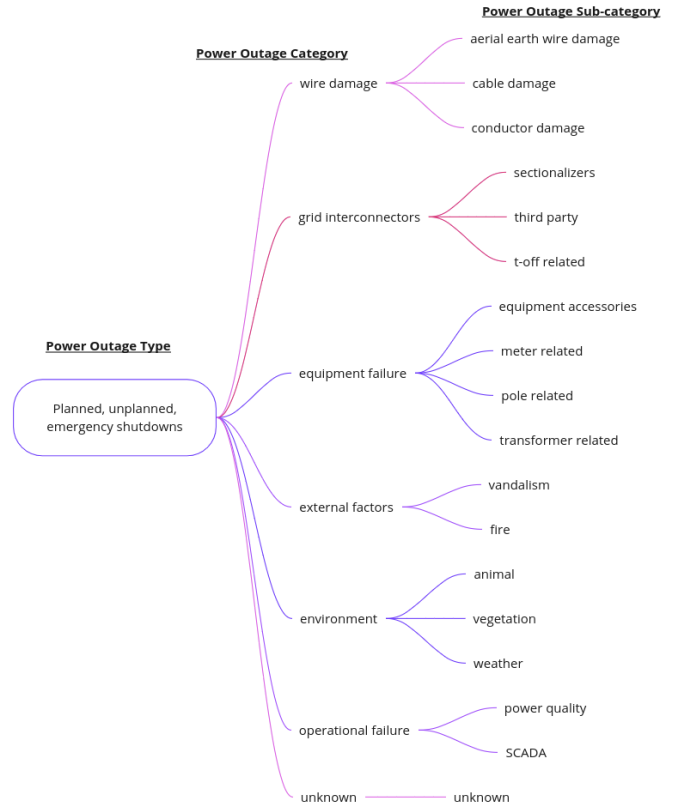


Fig. 2. Power outage root cause categorization

```

from setfit import SetFitModel, SetFitTrainer

model = SetFitModel.from_pretrained(
    "sentence-transformers/paraphrase-mpnet-base-v2",
    multi_target_strategy="one-vs-rest")

trainer = SetFitTrainer(
    model=model,
    train_dataset=train_dataset,
    eval_dataset=eval_dataset,
    metric=compute_metrics,
    num_iterations=5,
)

trainer.train()
metrics = trainer.evaluate()
model.save_pretrained("set-fit")
  
```

Fig. 3. Example code

signed for few-shot learning on sentence classification tasks, SetFit benefits from HuggingFace's easy-to-use interface for fine-tuning. While models from the BERT family (BERT, RoBERTa, DistilBERT, etc.) are highly regarded in the NLP community and provide similar capabilities, they are trained on longer texts. They may not suit our short description format as well.

B. Automated Labeling: Training classification model

As outlined in the data section, each outage in our dataset is accompanied by a descriptive account of its cause.

These descriptions vary in detail, ranging from explicit and comprehensive explanations of root causes to succinct and vague mentions, sometimes omitting the underlying cause altogether. To streamline the categorization process, the utility employs a set of predefined labels for outage causes, occasionally resorting to manual processing to assign these labels. The designated outage cause labels are depicted in Figure 2 in the Data section. Additionally, we introduce an "unknown" label category for descriptions that fail to provide any information about the outage's root cause(s).

We adopt a supervised learning approach utilizing the SetFit model introduced earlier to automate this classification process. We manually label a subset of our dataset, ensuring a representative sample from each label category. In total, 2452 cases are manually labeled for training and testing purposes, with an 80:20 split ratio.

IV. RESULTS

A. Automated Labeling: Model performance

As described in the methodology section, we developed a model to categorize outage descriptions into 18 different sub-categories. The model achieved a notable weighted average F1-score of 0.98 on the training dataset and an F1-score of 0.93 on the test dataset. The dataset comprises 215,387 outage description inputs; after excluding entries with missing feeder codes, this number narrows to 184,134 entries. The model classified 60% of the description entries as "unknown." This classification often includes descriptions like '*Fault; Reclosure Successful.*', which provide minimal insight into the outage's root cause. Such vague descriptions may result from technicians facing difficulty in determining the root cause, encountering outages of short duration, or experiencing challenges in proper documentation.

B. Power outage types

As mentioned in the data section, power outages are classified as planned, unplanned, or emergency shutdowns. This section examines the distribution of power outages across these types and the duration of each type.

Figure 4 shows that 82% of power outages are unplanned, 12% are emergency shutdowns, and 5.5% are planned shutdowns. The high proportion of unplanned outages highlights the grid's reliability challenges, which tend to reduce customer satisfaction and negatively impact economic activity.

Regarding the duration of outages, Figure 5 reveals that most unplanned and emergency outages are resolved within two hours. In contrast, most planned outages take six to ten hours to resolve.

C. Transient and Prolonged power outages

A transient power outage refers to a temporary interruption or loss of electrical power that lasts for a short duration. Unlike prolonged (non-transient) outages, which can last for hours or even days, transient outages typically occur for a

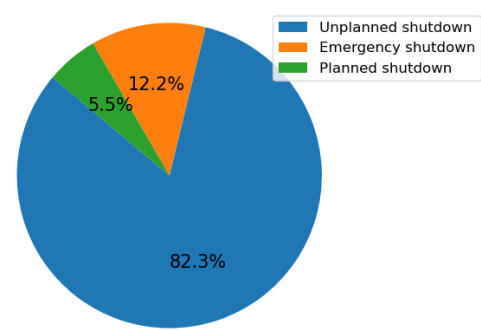


Fig. 4. Pie-chart showing power outage types by proportion of occurrence

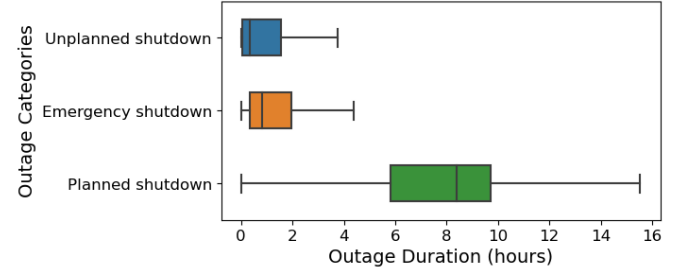


Fig. 5. Outage duration distribution of power outage types

brief period, often lasting only a few seconds or minutes. Using ERA standards, all power outages that last less than five minutes are labeled as transient power outages, and all outages lasting more than five minutes are non-transient power outages. In Figure 6, we show the proportion of outages that are transient and the proportion that are non-transient by power outage category.

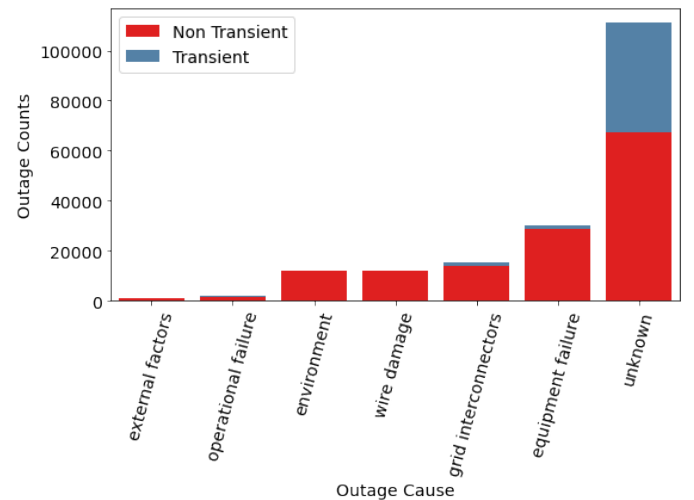


Fig. 6. Outage counts by category

In Figure 6, we also observe that our model cannot identify the root outage cause in most of our dataset's outage

description entries and attaches the “unknown” label to them. Interestingly, 40% of outages under the “unknown” label are transient, meaning they last less than 5 minutes. 8% of outages under the ‘grid interconnector’ category are transient, while less than 3% of the outages for each of the rest of the categories are transient. The temporary nature of many outages might explain why field technicians’ descriptions lack detail about the root causes. Many of these outages are likely resolved remotely, leaving no opportunity for technical teams to investigate further.

D. Power outage categories

Figure 7 illustrates the outage duration distributions by category type. We exclude transient and planned outages from our analysis to create this boxplot comparison. ERA applies the same standards when analyzing power outages. We observe that the median outage for each category lasts less than 2 hours.

Power outages under the “unknown” category have the smallest distribution of outage durations and the lowest median outage duration. We observe from the figure that half the outages in this category last less than 30 minutes before getting resolved. Outages in the ‘unknown’ category occur most frequently but get resolved most quickly.

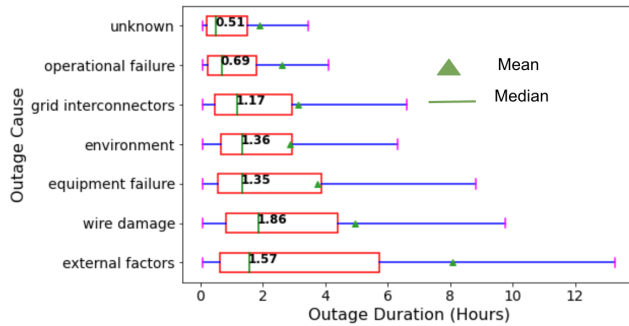


Fig. 7. Outage duration distribution by category

In the following paragraphs, we take a closer look at each of the remaining six categories.

1) *Equipment failure*: Equipment failure is the most prevalent cause of power outages. There are 29000 instances of power outages caused by equipment failure. The median outage resulting from equipment failure lasts for 1.35 hours. Half the outages in this category are caused by the failure of accessories such as broken jumpers (connectors), 30% of outages in this category arise from pole damage, 16% are due to transformer malfunction, and 2 % arise from challenges with meters.

2) *Grid interconnectors*: We defined grid interconnectors as other agents that connect to the Umeme grid. Examples include the main transmission company, UETCL (Uganda Electricity Transmission Company Limited), which connects

Umeme to generators. The other agents are distribution utilities that serve areas outside Umeme’s service territory. We also categorize interconnection points under this category, such as t-off points and sectionalizers. Grid interconnectors account for the second largest source of power outages. Faults originating from distribution utilities connected to Umeme’s grid can spread to Umeme’s grid. In some cases, protection at connection points is limited or absent, limiting Umeme’s capacity to isolate the affected distribution grid and cause a power outage on the Umeme grid.

Within this category, outages originating from agents interacting with the Umeme grid account for 56% of outages, with the remainder of outages resulting from failures at interconnection points.

3) *Wire damage*: Electric current and communications are transmitted using conductors and cables, which are mostly above ground. The exposed nature of this wire makes it especially susceptible to damage that results in power outages to either restore electricity transmission due to a damaged conducting wire or to restore communication due to damaged communication cables.

4) *Environment*: Environmental factors are the fourth most prevalent source of power outages. In our analysis, 60% of environment-related power outages resulted from bad weather and predominately rains, 36% were caused by vegetation, and 3% were identified as caused by animals. Mugenyi et al (2024). highlight the correlation between wet seasons and increased power outages in Uganda [23].

5) *Operational failure*: Power quality challenges refer to factors that affect the frequency and voltage of the power supply, such as under-frequency or overvoltage. Power quality challenges account for 88% of all operational failure power outages. The rest result from the failure of the SCADA (Supervisory Control and Data Acquisition) systems that are used to monitor grid operations remotely.

6) *External factors*: 60% of power outages resulting from external factors are caused by fires. The remaining 40% are attributable to cases of vandalism. Even though cases caused by vandalism and fire have the least frequency of occurrence, the data shows that they have the widest variation in resolution times, as shown in Figure 7. Speculatively, the significant amount of damage resulting from such cases could explain why many cases take much longer to resolve.

V. CONCLUSION & RECOMMENDATIONS

Through a collaboration with the Electricity Regulatory Authority in Uganda, we obtained a detailed data set of power outages recorded from 2015 to 2022. Each entry in the dataset includes a free-form description detailing the nature of the outage.

We employed SetFit, an advanced natural language processing (NLP) model, to automate the categorization of these outage descriptions provided by technicians. By training this model, we were able to assign specific root cause labels to

each description, facilitating a comprehensive analysis of the predominant causes of power outages in Uganda.

Our findings indicate that equipment failure is the leading cause of power outages. External factors such as vandalism and fires also result in significantly prolonged outages. These insights highlight the urgent need to reassess and enhance maintenance practices to reduce the frequency of outages. Moreover, the extended resolution times for outages caused by vandalism and fires suggest specific areas where targeted improvements could further improve the reliability of Uganda's power supply.

This research demonstrates the effectiveness of NLP in deriving actionable insights from extensive data sets, providing a valuable framework for other regions to identify and mitigate their power outage issues. By leveraging such technology, stakeholders can better understand and address the root causes of power disruptions, ultimately leading to more stable and efficient power systems.

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